

Sustainability and Environmental Management Systems

What is Sustainability?

- Considerable and intense utilization of natural resources and organizational resources in a judicious and responsible manner.
- Developing choices that reflect on the long-term effectiveness and not on short-term outcomes.
- Allowing the future generation to benefit positively when the action of the present generation is in question.



Figure: Sustainability in business.
Source: United Nations

Main Areas of Sustainability.

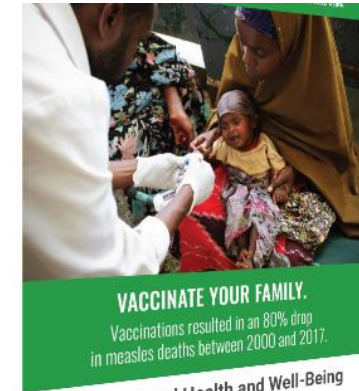
- Environmental responsibility, such as safeguarding the natural resources and minimization of pollution.
- Social responsibility including being fair to employees and giving back to communities.
- Economic responsibility, which aims at maintaining profitability in the long run.



Goal 1: No Poverty



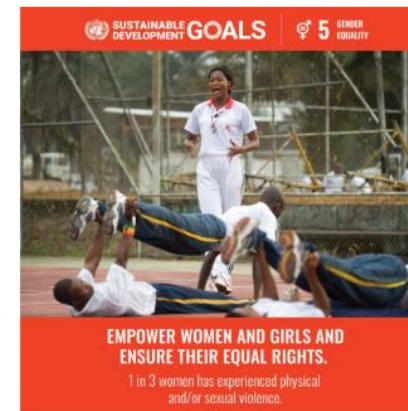
Goal 2: Zero Hunger



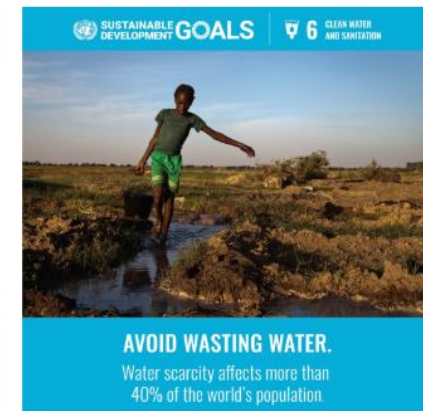
Goal 3: Good Health and Well-Being



Goal 4: Quality Education



Goal 5: Gender Equality



Goal 6: Clean Water and Sanitation

Figure: Areas of Sustainability
Source: UN

Sustainability in Business Setup.

- Less carbon emissions and environmental impact reduction from the operations.
- Enhanced use of energy, water, and raw materials in the organization.
- Reducing the amount of waste production and eliminating pollution where necessary.

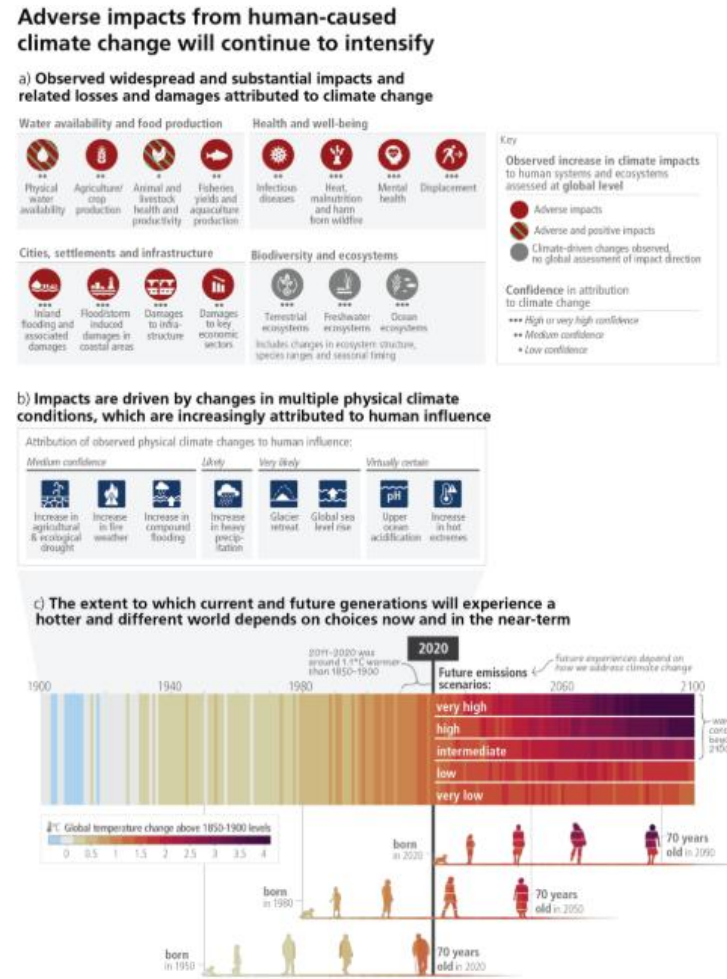


Figure: Carbon effects trends
Source: IPCC (2023)

Accountability and Assumptions.

- Adhering to the laws and regulations of the industry in order to escape fines.
- Responsible management of the supply chains in order to maintain ethical sourcing and production.
- Delivering customer, investor, and other stakeholder expectations.

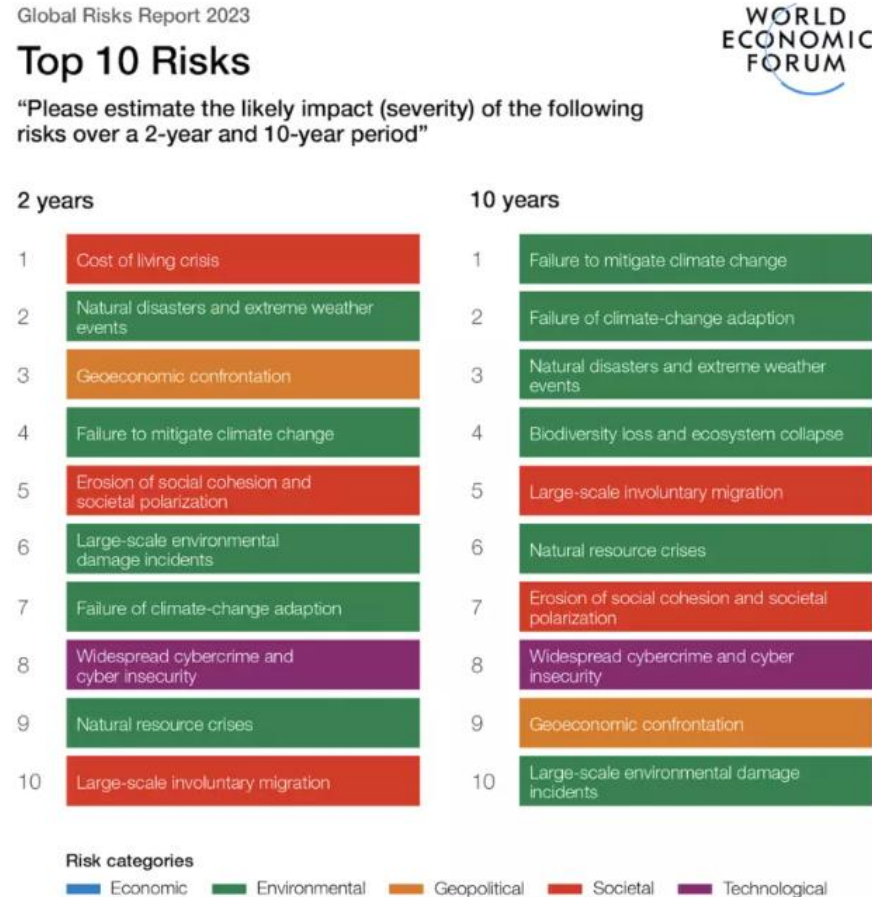


Figure: Sustainable supply chain

Source: World Economic Forum (2023)

Environmental Management System, what is it?

- An organized method that organizations employ in order to deal with the impacts of the environment.
- Plans the sustainability-related operations in an organizational manner.
- Incorporates environmental considerations in the normal business decisions.

Figure 1 — Relationship between PDCA and the framework in this International Standard

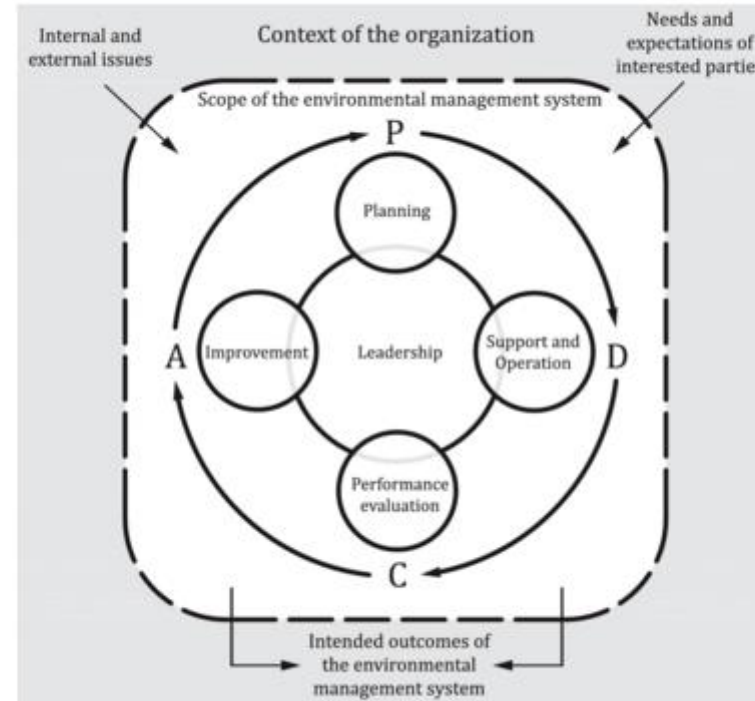


Figure: EMS Structure
Source: ISO 1400

Purpose of an EMS

- Assists the organizations in observing the impacts of their operations on the environment.
- Enhances productivity by establishing where resources are squandered.
- Makes sure that every legal and regulatory requirement is being addressed.

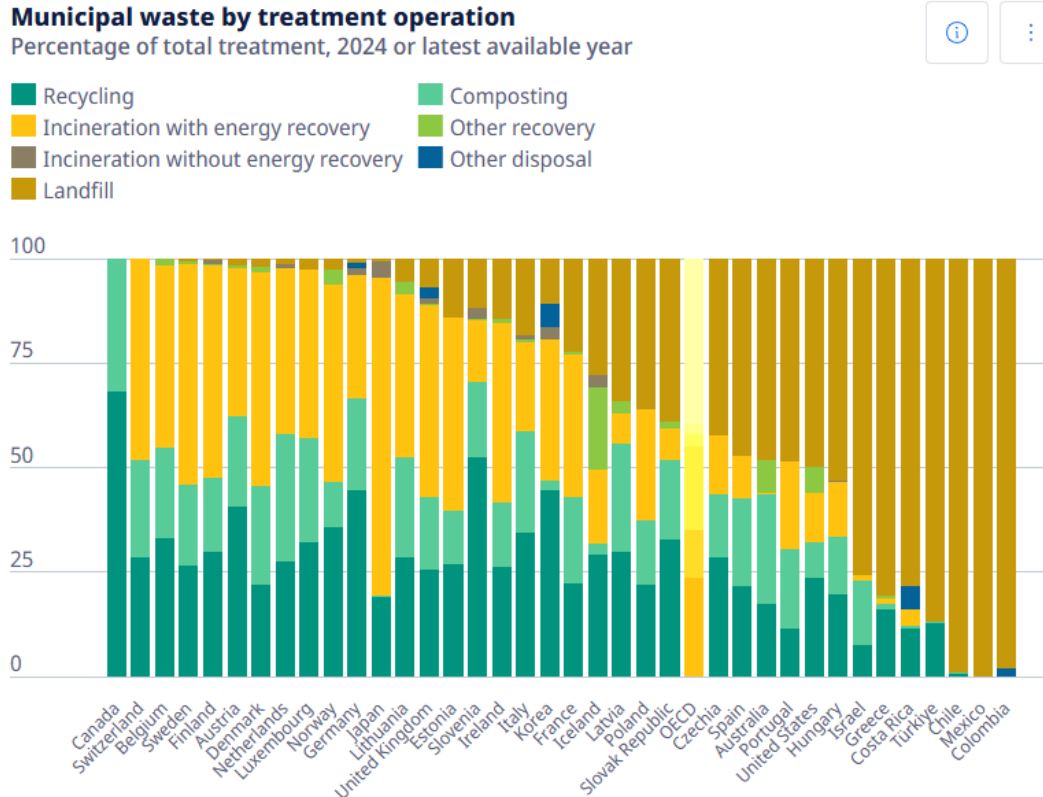


Figure: Environmental Performance graph
Source: OECD (2023)

How EMS Works

- Planning phase—identification of environmental impacts and setting of objectives.
- Implementation stage where control and processes are realized.
- Constellation stage where the performance is checked and assessed.
- Improvement stage whereby decisions are made because of the results.

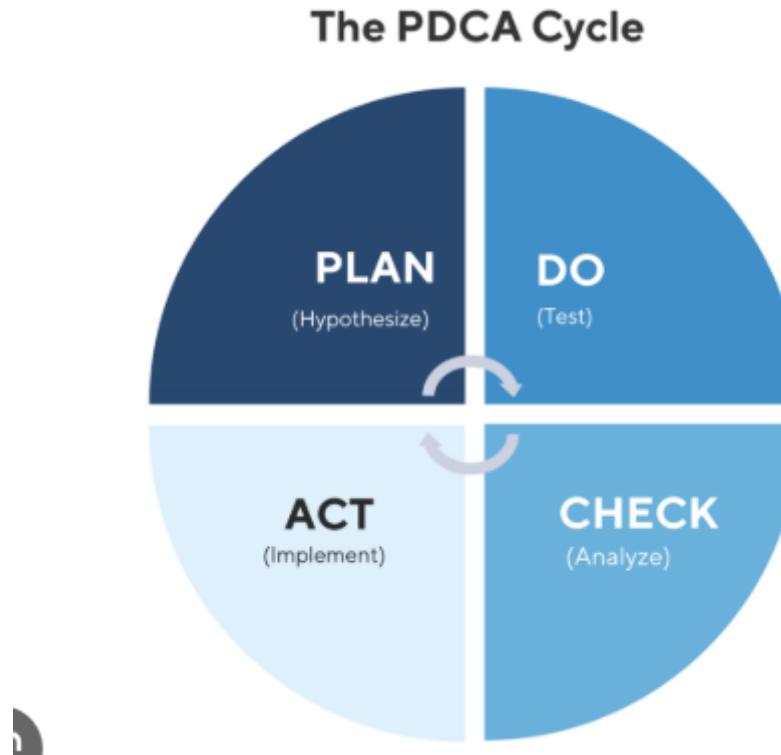


Figure: PDCA Cycle
Source: ISO (2024)

Connection between sustainability and EMS.

- Sustainability offers the general meaning and aim of the environmental action.
- EMS puts sustainability goals into practice and process.
- They are taken together, both to provide long-term thinking and day-to-day management.

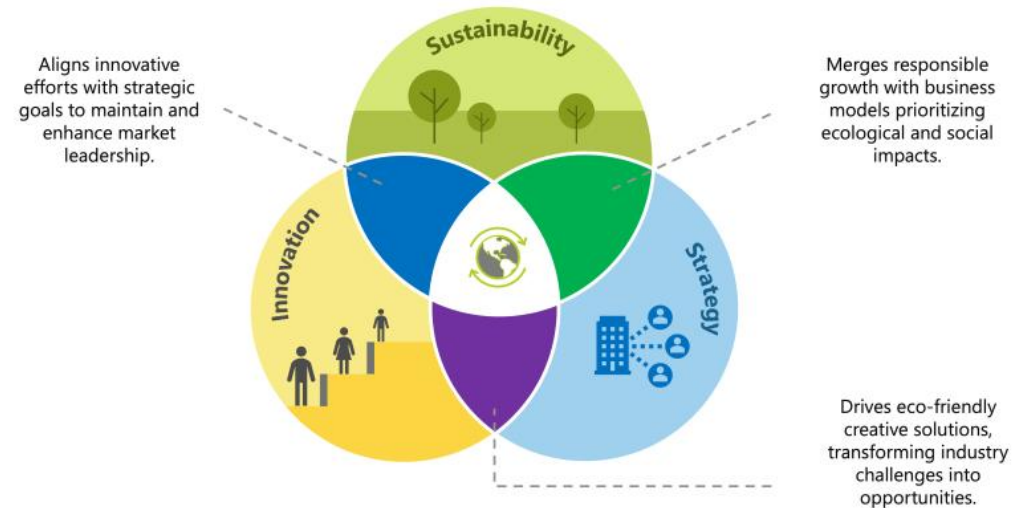


Figure: Sustainability, Innovation and Strategy
Source: Prospectus (2024)

The way sustainability designs EMS.

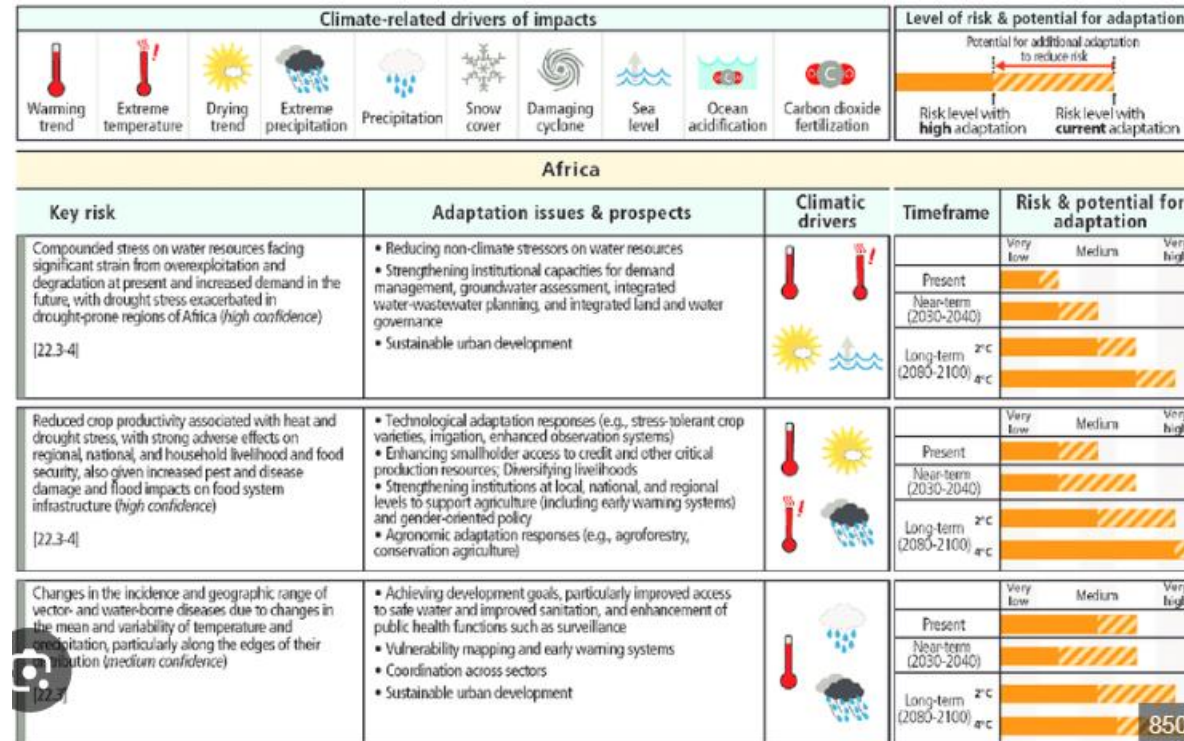
- Affects environmental policy creation processes in the organization.
- Promotes long-term goal setting rather than short-term solutions.
- Facilitates participation of various stakeholders within the organization.



Risk Management Framework
Source: WEF (2024)

The Need for Sustainability in Organizations.

- Specializing in more environmental regulations that organizations have to abide by.
- The increasing threats posed by climate change that can disrupt operation.
- Low supply of natural resources for long-term growth.

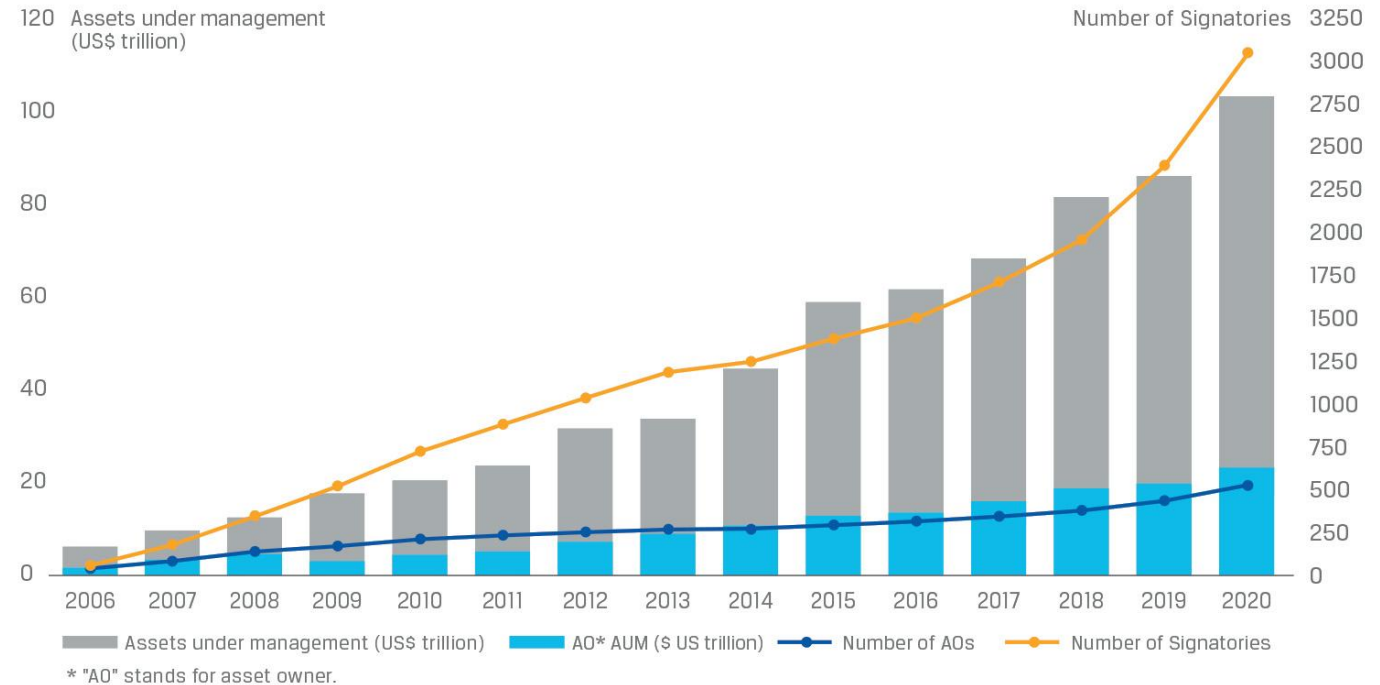


Climate Risk Assessment

Source: IPCC (2023)

Pressures on Organizations Which are external.

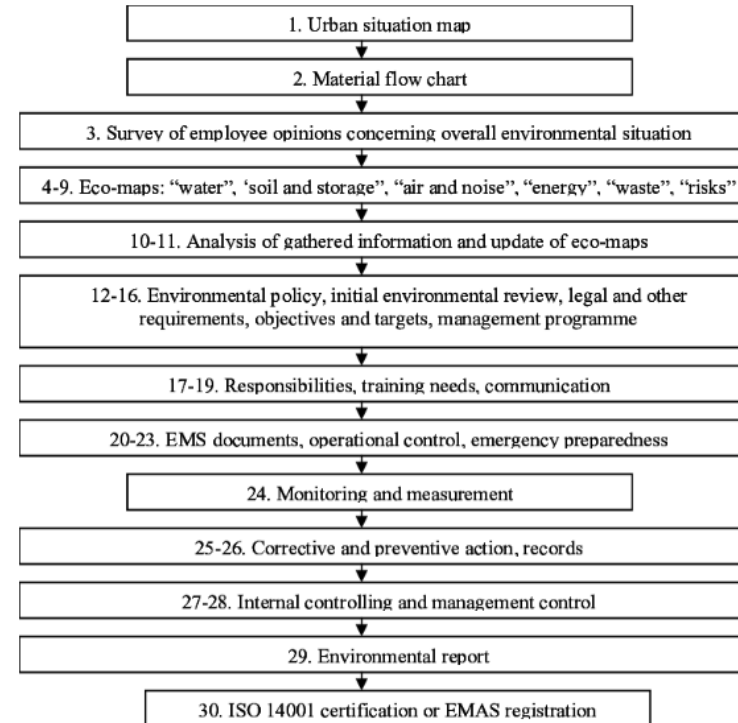
- Investors are increasingly concentrating on responsible business in terms of the environment.
- Customer: Customers like organizations that practice sustainability.
- There is a growing competition with rising implementation of sustainability strategies by more companies.



ESG Growth Chart
Source : PwC (2024)

The support of EMS to sustainability.

- Gives good procedures to control environmental responsibilities.
- Action: Measures performance in a quantifiable way to monitor the progress.
- Incorporates sustainability into operations and making decisions.

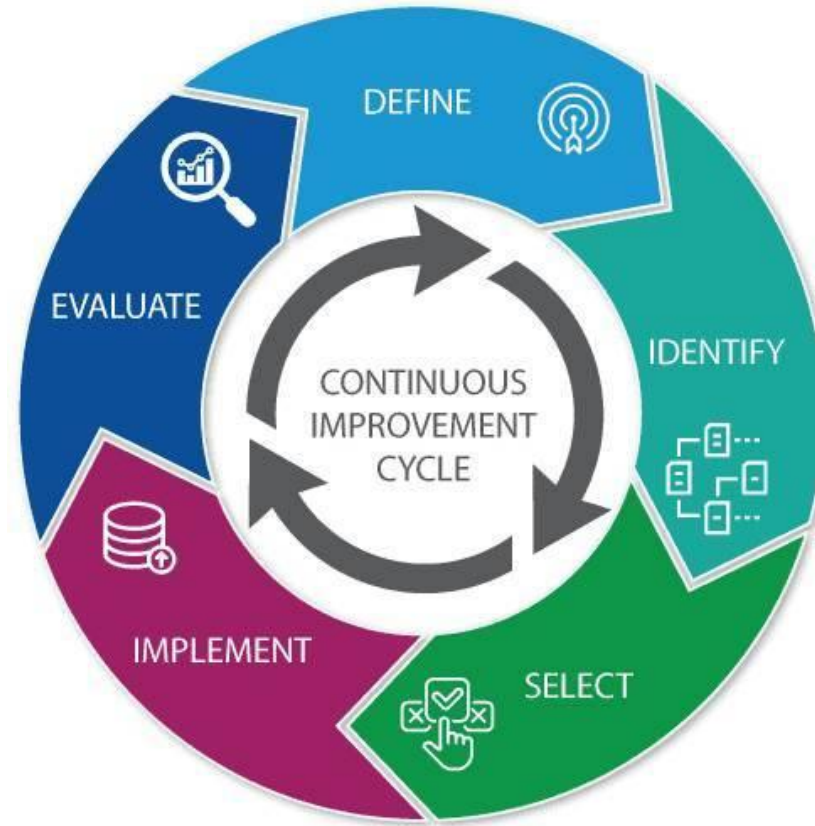


30 steps for EMS implementation in SME (special/ standard forms are used for each step)

EMS implementation flow
Source: ISO (2024)

Continuous Improvement.

- Laying down precise and quantifiable environmental targets of the organization.
- On a regular basis, performance should be reviewed to determine areas of weakness.
- Continuing to make changes to better efficiency and less impact according to Deloitte (2023) .



Continuous Improvement Cycle
Source: Deloitte (2023)

EMS Operational Benefits.

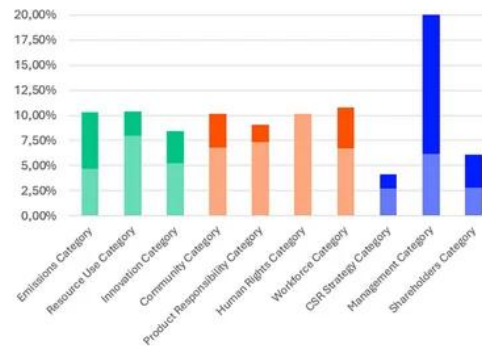
- Cuts back on the operation expenses through efficiency and wasting.
- Enhances better utilization of energy, water, and other resources.
- Reduces environmental accidents and fines.



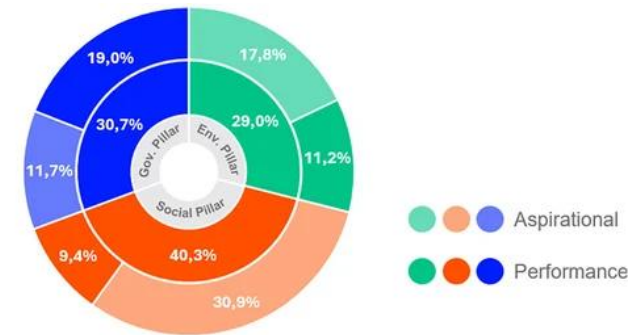
Cost Saving from Sustainability
Source: OECD (2022)

Strategic Advantages to EMS.

- Enhances the reputation and the image of the organization.
- Gains customer trust, investor trust, and regulatory trust.
- Promotes creativity and creation of new opportunities.



(a) ESG Average Category Weights: the histogram represents the contribution of each category to the final ESG score according to Table 6. Colors are chosen according to pillar differentiation, and their intensity underlines the *aspirational/performance* metrics nature.



(b) ESG Average pillar Weights: the chart represents the overall contribution of each pillar to the final ESG score according to Table 6. Color intensity underlines the *aspirational/performance* proportion for each pillar.

ESG Score

Source: Deloitte (2023)

Conclusion

- Sustainability is emerging as a key to success on a long-term basis.
- EMS is an organization that has a systematic means of attaining sustainability goals.
- Early-acting organizations have chances of succeeding in the future.



Sustainability Roadmap

Source: IEA (2023)

References

- Deloitte (2023). *The sustainability advantage: Driving long-term business performance*. Deloitte Insights.
- IEA. *Energy Efficiency 2023*. IEA
- ISO (2024). *ISO 14001: Environmental management systems – Requirements with guidance for use*. ISO.
- United Nations (2023). *The sustainable development goals report 2023*. United Nations.
- PwC (2024). *Global ESG investor survey 2024*. PwC